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Lab 02 - First Verilog

In this lab, you've learned how to do an initial and simple design in Verilog to learn the Vivado tooling and process involved in RTL/FPGA design.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

In this first lab primarily, we learned about starting a new project in Vivado and testing through a simple Basys 3 board. The lab required an assignment from switch zero to LED zero and then exploring Vivado. Through Vivado we were able to implement the code to get the LED light to turn on.

Lab Questions

1 - Describe the stages of building a Verilog project in Vivado.

After opening Vivado and clicking 'Create Project', we need to designate a project directory and leave create subdirectory checked. From here we create the RTL project without specifying sources until the following screens. Now we add in our source files containing the written Verilog code followed by the constraints document to connect to the board. The project has now been created. Now we can simulate to check behavior and then synthesize it into a hardware list. The final steps are to implement onto the FGPA with placement and routing. The final steps are to generate the bit stream and then program the board.

2 - What is the value in looking at the elaborated design schematic?

The elaborated design shows a visual representation of the LED with respect to switches. In the case of this lab, we can clearly see the array of switches [15:0] connected to the array of LED [15:0]. This view could also be useful when working with multiple logic gates or advanced boards.

3 - Why should we simulate our designs frequently? What does the simulation do?

By simulating frequently, we can test the design before implementing it to the board. In cases of complicated circuits, we can see where errors come from. We can see conflicts between parts of the system especially since the code executes all at once.

Code Submission

Upload a .zip of all your code or a public repository on GitHub.